

JCB E-Sell DBMS Integration Structure/Runbook

JCB E-Sell and DBMS Data Integration Runbook

Document Information

Field	Details
Document Name	JCB E-Sell and DBMS Data Integration Runbook
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Application	JCB E-Sell & JCB DBMS Integration
Integration Type	SFTP-Based XML Data Exchange

Version Control

Name	Author	Date	Remarks	Version
JCB E-Sell and DBMS Data Integration Runbook	Aditya Mishra	01/11/2018	Created as per RFP and shared docs	Initial Draft

1. Purpose

This runbook provides operational procedures for monitoring, executing, validating, and troubleshooting the data integration process between the JCB E-Sell platform and the JCB DBMS system.

The document is intended for application support teams, infrastructure teams, database administrators, and operations personnel responsible for maintaining the integration environment.

2. Scope

This runbook covers:

- Data extraction from JCB E-Sell (MRM Server)
- XML generation and encryption
- SFTP file transfer
- XML decryption and database loading
- Dealer application synchronization
- Sales data synchronization back to JCB E-Sell
- Monitoring and alert management
- Troubleshooting and recovery procedures

3. System Overview

The integration enables automated data exchange between the JCB E-Sell platform and the JCB DBMS system.

Objectives

- Eliminate manual order number entry.
- Remove Excel-based sales closure activities.
- Improve data accuracy.
- Reduce operational effort.
- Enable automated synchronization of dealer sales information.

4. Environment Details

Refer below overview of the infrastructure, servers, databases, network configurations, and deployment environments supporting the JCB E-Sell and DBMS integration solution, ensuring smooth operation, monitoring, and maintenance of the application ecosystem.

Source System

The **Source System** refers to the originating application or platform from which data is generated and transmitted to downstream systems as part of the integration process. In the JCB E-Sell and DBMS integration architecture, the JCB E-Sell platform acts as the primary source system for dealer orders, customer purchase information, and sales transaction data that is shared with external systems for further processing and invoicing.

Component	Details
Application	JCB E-Sell
Hosting	MRM Server
Database	SQL Server
Output Format	XML

Destination System

The **Destination System** refers to the target application, database, or platform that receives and processes data transmitted from the source system. In the JCB E-Sell and DBMS integration solution, the JCB DBMS/JFC Server acts as the primary destination system, where order, dealer, and sales information received from the JCB E-Sell platform is decrypted, validated, and loaded for invoicing, sales processing, and downstream business operations.

Component	Details
Application	JCB DBMS
Hosting	JFC Server (Delloy 01)
Database	SQL Server

Transport Layer

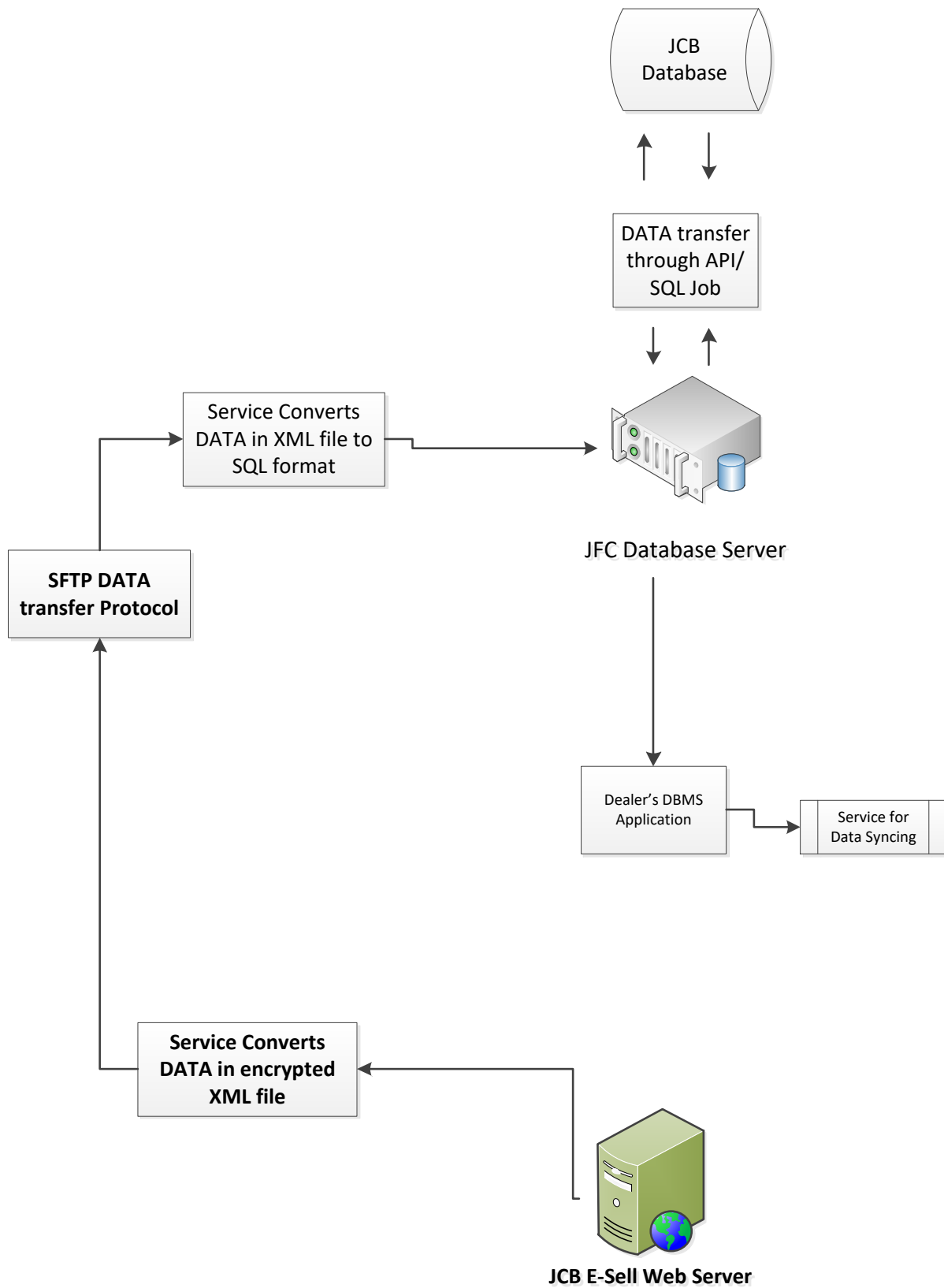
The **Transport Layer** defines the communication mechanism used to securely transfer data between the source and destination systems. In the JCB E-Sell and DBMS integration solution, data is transmitted through the **Secure File Transfer Protocol (SFTP)** over an encrypted channel, ensuring secure, reliable, and controlled movement of XML files between the MRM server and the JFC/DBMS server while protecting sensitive business data from unauthorized access or interception.

Component	Details
Protocol	SFTP
Encryption	AES-256
Authentication	Private Key

5. Integration Workflow

The integration process consists of the following stages:

1. Dealer order data is generated within JCB E-Sell.
2. XML file is created by the integration service.
3. XML file is encrypted using AES-256.
4. Scheduler pushes encrypted file to JFC Server through SFTP.
5. File is received by JFC integration service.
6. XML file is decrypted.
7. Data is converted into SQL-compatible format.
8. Data is inserted into JFC database.
9. Dealer DBMS application synchronizes data.
10. Dealer performs invoicing using available order data.
11. Sales information is transferred back to JCB E-Sell.
12. Success or failure notifications are generated.



JCB E-Sell and DBMS Integration Architecture

6. Daily Operations Checklist

Perform the following checks daily.

Activity	Frequency	Owner
Verify Integration Service Status	Daily	Application Support
Verify Scheduler Execution	Daily	Operations Team
Verify SFTP Connectivity	Daily	Infrastructure Team
Verify XML File Generation	Daily	Application Support
Verify File Encryption	Daily	Application Support
Verify Database Upload Success	Daily	DBA Team
Verify Email Notifications	Daily	Support Team

7. Operational Procedure

Step 1: Verify Integration Service

Objective

Ensure that XML generation service is operational.

Validation

- Service status should be Running.
- No service errors should be present in logs.

Expected Result

XML generation service available.

Step 2: Verify XML File Generation

Objective

Confirm successful XML generation.

Validation

- XML file exists in source directory.
- File timestamp matches current schedule.

Expected Result

New XML file generated successfully.

Step 3: Verify Encryption

Objective

Ensure generated XML files are encrypted.

Validation

- Encrypted file available.
- File extension matches configured standard.

Expected Result

Encrypted XML package ready for transfer.

Step 4: Verify SFTP Transfer

Objective

Confirm successful file transmission.

Validation

- File present on JFC server.
- Transfer logs show Success.

Expected Result

File successfully delivered.

Step 5: Verify Decryption Process

Objective

Ensure files are decrypted successfully.

Validation

- No decryption errors.
- XML structure readable.

Expected Result

File successfully decrypted.

Step 6: Verify Database Upload

Objective

Confirm data loading into JFC database.

Validation

- Record counts match source.
- No SQL exceptions.

Expected Result

Data available in destination tables.

Step 7: Verify Dealer Application Sync

Objective

Confirm dealer application receives updated orders.

Validation

- Order appears in dealer popup.
- Dealer can select order for invoicing.

Expected Result

Order visible to dealer.

8. Monitoring

Monitor the following components continuously.

Component	Monitoring Item
Integration Service	Service Availability
Scheduler	Job Completion
SFTP	Connectivity
Database	Upload Success
Email Service	Notification Delivery
Disk Space	Storage Utilization

9. Alert Management

Success Notification

An automated email should be generated when:

- File transfer completes successfully.
- Database upload completes successfully.

Expected Subject

JCB Integration Success - File Processed Successfully

Failure Notification

An automated email should be generated when:

- XML generation fails.
- Encryption fails.
- SFTP transfer fails.
- Database upload fails.
- Dealer sync fails.

Expected Subject

JCB Integration Failure - Immediate Attention Required

10. Troubleshooting Guide

Issue 1: XML File Not Generated

Symptoms

- No XML file available.
- Scheduler completed successfully.

Resolution

1. Verify integration service status.
2. Review application logs.
3. Restart integration service.
4. Re-execute scheduler.

Issue 2: SFTP Transfer Failure

Symptoms

- File missing on destination server.
- Transfer log contains connection error.

Resolution

1. Verify SFTP credentials.
2. Verify firewall access.
3. Verify SSL port accessibility.
4. Retry transfer.

Issue 3: Decryption Failure

Symptoms

- XML file cannot be read.
- Decryption service reports error.

Resolution

1. Verify encryption key.
2. Verify key configuration.
3. Re-transfer source file.

Issue 4: Database Upload Failure

Symptoms

- Records not available in JFC database.
- SQL exception generated.

Resolution

1. Review SQL logs.
2. Validate XML structure.
3. Verify database connectivity.
4. Reprocess failed file.

Issue 5: Dealer Cannot View Orders

Symptoms

- Dealer popup is empty.
- Orders exist in JFC database.

Resolution

1. Verify dealer synchronization service.
2. Verify API/service connectivity.
3. Verify dealer mapping records.

11. Recovery Procedure

If a transfer fails:

1. Identify failed stage.
2. Correct underlying issue.
3. Reprocess affected file.
4. Validate successful upload.
5. Confirm dealer visibility.
6. Notify stakeholders.

12. Escalation Matrix

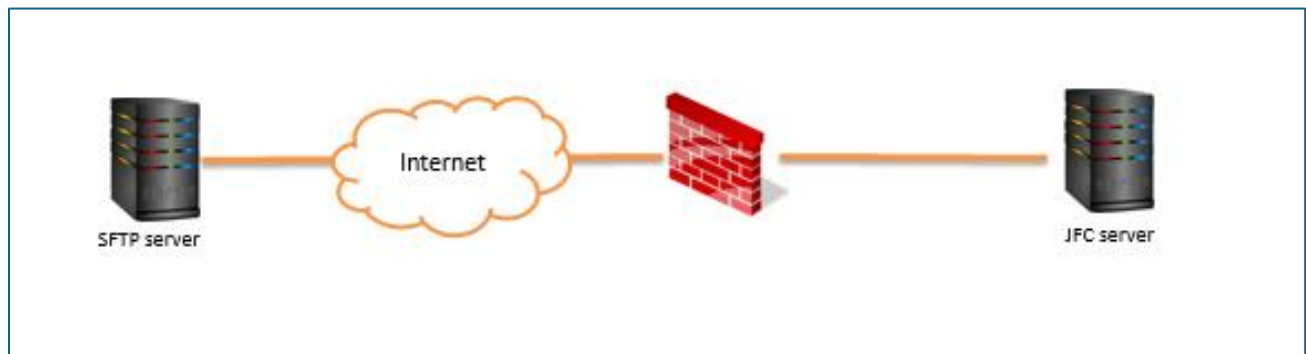
Severity	Description	Escalation Team
P1	Complete integration outage	Infrastructure + Application Support
P2	Data transfer delay	Application Support
P3	Single dealer issue	Functional Support Team
P4	Reporting issue	Service Desk

13. Security Controls

The following controls must remain enabled:

- SFTP secure communication
- AES-256 encryption

- Firewall restrictions
- SSL-based connectivity
- Audit logging
- Access control
- Secure key management

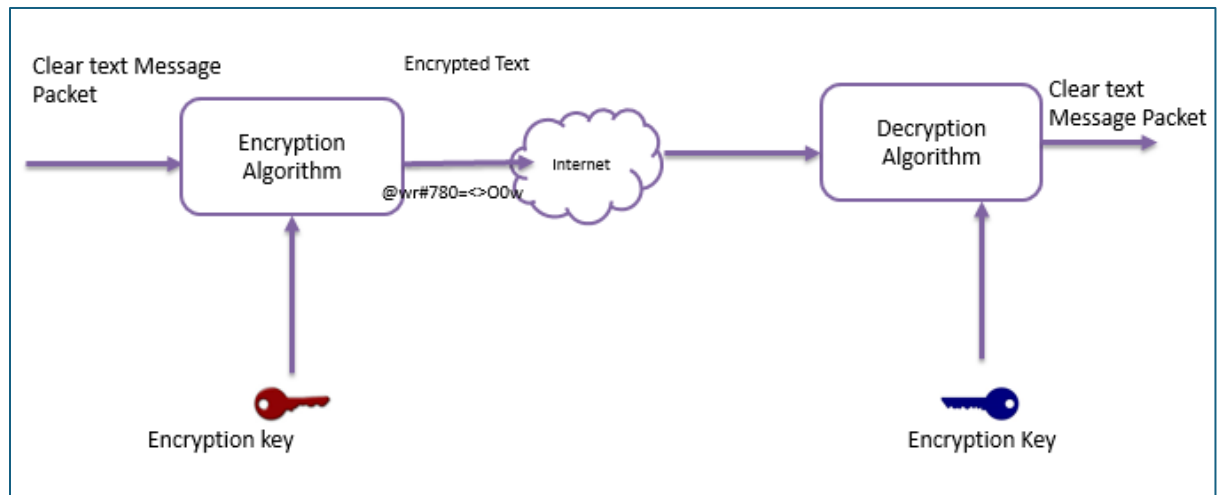


14. Validation Checklist

After every successful run:

- XML generated
- File encrypted
- File transferred
- File decrypted
- Data loaded
- Dealer sync completed
- Sales sync completed
- Notification generated

All checkpoints must pass before the integration cycle is considered complete.



Glossary

Term	Definition
AES-256	Advanced Encryption Standard (256-bit) used to encrypt XML files during data transmission between systems.
API	Application Programming Interface used for system-to-system communication and data exchange.
DBMS	Dealer Business Management System used by dealers for invoicing, order processing, and sales management activities.
Dealer	Authorized JCB business partner responsible for processing customer orders, invoicing, and product delivery.
DSA	Dealer Sales Associate who can place orders on behalf of customers through the E-Sell platform.
Encrypted XML File	XML data file secured using encryption before being transferred between servers.
E-Sell Platform	JCB's online parts ordering portal where customers and DSAs can browse and purchase genuine JCB parts.
File Retention Period	The duration for which transferred files are stored before automatic deletion. In the proposed solution, files are retained for 15 days.
FTP	File Transfer Protocol used for transferring files between systems. SFTP is the secure version of FTP.
Integration Service	Background application responsible for generating, encrypting, transmitting, decrypting, and processing data files.
Invoice	Commercial document generated by dealers in the DBMS system after processing an order.
JCB Central Database	Centralized JCB repository that stores consolidated business, sales, and dealer information.
JFC Server (Delloy 01)	JCB-hosted server responsible for receiving, decrypting, processing, and storing data received from the MRM environment.
MRM Server	Application and database infrastructure hosting the JCB E-Sell platform and integration services.
Offline Sales	Sales transactions processed outside the online payment flow, typically managed through dealers or DSAs.
Online Sales	Sales transactions completed through the E-Sell platform using an integrated payment gateway.
Order Number	Unique identifier assigned to every customer order placed through the JCB E-Sell platform.
Payment Gateway	Third-party service integrated with E-Sell to facilitate online payments.
Pop-up Order Selection	Proposed enhancement allowing dealers to select order numbers directly from a system-generated list instead of manual entry.
Private Key	Cryptographic key used for encrypting and decrypting XML files during secure data transfer.
Scheduler Service	Automated background process that triggers file generation, transmission, and cleanup activities at predefined intervals.
SFTP	Secure File Transfer Protocol used for encrypted file transmission between MRM and JFC servers.

Source System	System that originates and sends data. In this integration, the JCB E-Sell platform acts as the source system.
Destination System	System that receives and processes data. In this integration, the JFC/DBMS environment acts as the destination system.
SQL Job	Scheduled database task used to automate data synchronization and transfer between systems.
SQL Server	Relational database platform used to store and manage business data in the JCB ecosystem.
Transaction Log	Record containing details of file transfers, processing events, and system activities for auditing purposes.
Transport Layer	Communication layer responsible for securely transferring data between source and destination systems using SFTP.
XML	Extensible Markup Language used as the standard data exchange format between JCB E-Sell and DBMS systems.
XML Conversion Service	Service that converts database records into XML format before transmission and converts XML back into database records upon receipt.
Data Synchronization	Automated process of keeping information consistent between JCB E-Sell, DBMS, and Central Database systems.
Audit Trail	Historical record of all data transfer, processing, and synchronization activities for compliance and troubleshooting.
Firewall Whitelisting	Security configuration that allows only approved IP addresses and ports to communicate with JCB servers.
SSL Port	Secure communication port used to establish encrypted connections between systems during file transfer.

Abbreviations

Acronym	Full Form
API	Application Programming Interface
AES	Advanced Encryption Standard
DBMS	Dealer Business Management System
DSA	Dealer Sales Associate
FTP	File Transfer Protocol
JFC	JCB Finance Corporation / JCB Central Infrastructure (as applicable)
MRM	McCann Relationship Marketing
SFTP	Secure File Transfer Protocol
SQL	Structured Query Language
SSL	Secure Sockets Layer
XML	Extensible Markup Language